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Repelling Polysulfide Ions by Boron Nitride Nanosheets Coated Separators in Lithium-Sulfur Batteries

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Abstract

Migration of cycling intermediates (polysulfides) from cathode to anode through separator is one of main problems in current lithium-sulfur batteries that deteriorates cycling performance of the cell. Here we report a multifunctional separator, which is constructed by incorporation of functionalized boron nitride nanosheets with negative charged groups onto a commercial Celgard separator. The boron nitride separator is capable to prevent polysulfide migration through the separator effectively due to strong ion-repelling of negatively charged polysulfides by the

negatively-charged boron nitride nanosheets. The lithium-sulfur cell with a boron nitride separator exhibits an excellent long-term cycling stability up to 1000 cycles and a high capacity of 718 mA h g⁻¹ at a very high current of 7C (1.18 A g⁻¹). Such an excellent high rate capacity and long-term stability of the cell are certainly contributed by the boron nitride separators that inhibit internal redox shuttle and increase the diffusion coefficient of lithium ions.

Lithium-Sulfur (Li-S) batteries exhibit a very high theoretical specific capacity of 1675 mA h g⁻¹ and high energy density of 2650 W h Kg⁻¹ and are thus considered as next-generation lithium (Li) batteries.^{1, 2} Current battery devices still face some serious problems and related unsatisfied electrochemical performance. One major problem is the dissolution of intermediate products formed during charge-discharge process, which is the so-called “shuttle effect”, causing severe capacity decay and lower Coulombic efficiency of Li-S batteries.^{3,4} To overcome this problem, a significant research has been focused on the fabrication of new electrodes with a higher sulfur loading as well as incorporation of an interlayer between sulfur cathode and separator, such as carbon-sulfur composites,^{5,6} carbon nanotube paper,⁷ carbonized eggshell membrane,⁸ and functionalized boron nitride nanosheet combined graphene,⁹ which reduce the shuttle effect and significantly enhance rate performance and cycling life of batteries. Another approach to constrain shuttle effect of the Li-S battery is to modify the composition of the electrolyte. Certain electrolytes can prevent the dissolution of polysulfides (PSs), thus reducing the shuttling of PSs,¹⁰ however, ion conductivity in these electrolytes is sometime sacrificed, which leads to a low electrochemical performance of the battery. Recently, tailoring the physical and chemical properties of the

separator is demonstrated to be a viable way for further improving the performance of Li-S batteries.¹¹

To prevent PS shuttle effect efficiently, development of new separators with graphene oxide (GO),^{12,13} black-phosphorus,¹⁴ metal-organic framework (MOF)/GOs,¹⁵⁻¹⁷ and metal oxides (such as Al₂O₃¹⁸ and TiO₂¹⁹) have already been tried. It is demonstrated that the shuttle phenomenon in Li-S batteries can be effectively inhibited with these new separators through either physical trapping or chemical interactions of the separator with polysulfide species. Nevertheless, most of these modifications ultimately impede electron transport or interrupt paths for Li⁺ movement, thus leading to low sulfur utilization and rate capability. Therefore, development of new separators which not only inhibits PS shuttle effect but also increase electron transport speed as well as Li⁺ diffusion rate is a challenge.

Among different inorganic materials, boron nitride nanosheets (BNNSs), as an analogue of graphene, might be better material to solve this issue as BNNSs have been proven to promise many new properties,²⁰⁻²⁴ and a wide range of applications such as composite reinforcement, hydrogen storage, and adsorption of dyes, proteins, organic solvents, and oils due to high surface area, strong electrostatic attraction and non-covalent interaction.²⁵⁻³¹ Moreover, we have recently demonstrated that functionalized boron nitride (FBN) nanosheets with amino groups can be used for the fabrication of a graphene-FBN nanosheet interlayer that can protect sulfur cathode by adsorbing PS anions in Li-S batteries.⁹ Inspired by this previous work, herein, we develop a multifunctional thin separator by coating a layer of FBN nanosheets with CO₃²⁻ groups on a Celgard 2400 separator. The multifunctional FBN separator prevents PS shuttle effect dramatically and enhances Li ion diffusion significantly due to the incorporation of BNNSs with strong negative functional groups onto a Celgard separator. The cell with the FBN separator exhibits a high reversible

capacity, an attractive rate capability, and a more stable cycle life (without compromising its beneficial attributes) than that of the cell with a Celgard separator.

Results and discussion

A schematic configuration of a Li-S cell with an FBN separator is displayed in **Figure 1a**. The FBNs with CO_3^{2-} groups were prepared according to our previous report (see experimental in detail).³⁰ The scanning electron microscopy (SEM) image of the coating layer with FBN content of $0.21 \pm 0.01 \text{ mg cm}^{-2}$ in Figure 1b shows a cross-section of the coating layer on a separator. The low magnification SEM image in Figure S1 and high magnification SEM image shows that FBN nanosheets were coated on some surface of the Celgard 2400 separator without totally blocking Li ion transport channels (Figure 1c). The presence of CO_3^{2-} groups in FBN nanosheets was confirmed by Raman spectroscopy analysis (Figure 1d). Moreover, zeta potential measurement demonstrates that FBN nanosheets exhibit high negative potential value of $-44 \pm 3 \text{ mV}$, which is beneficial for ion-repelling of other negative ions.³²⁻³⁵

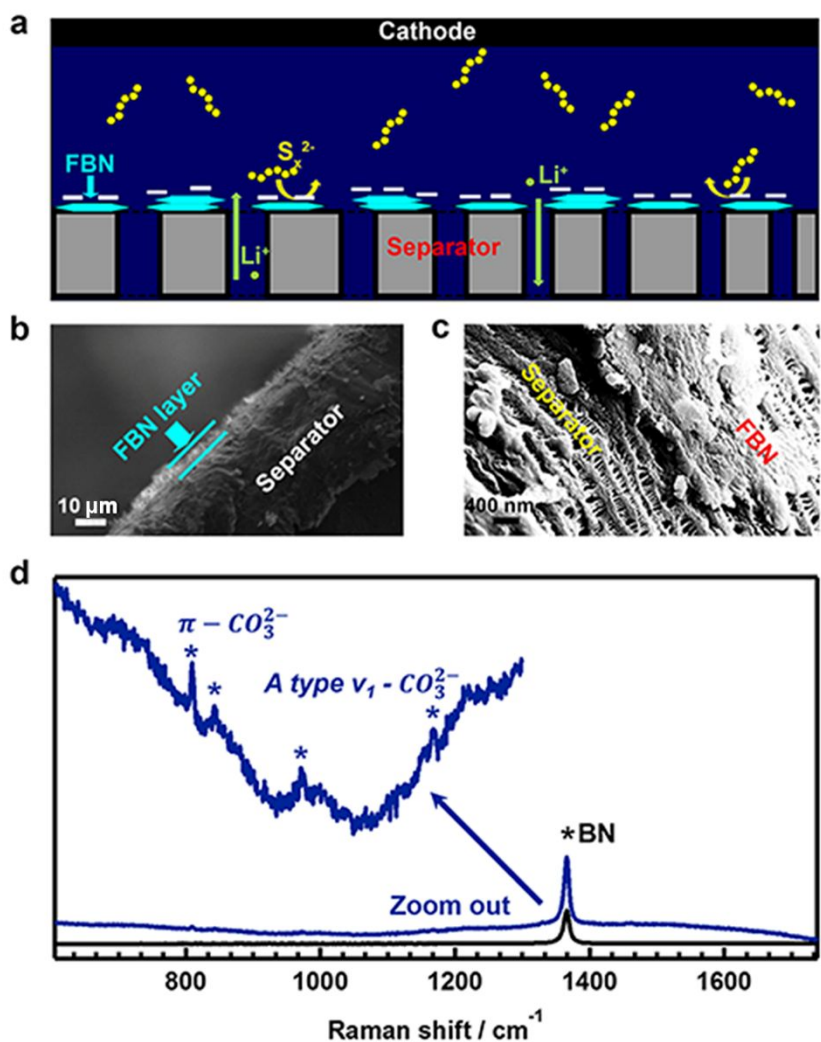


Figure 1 Structure of the Li-S cell with an FBN separator. **a.** schematic configuration of a sandwich-like Li-S cell with a FBN separator; **b.** SEM image of the cross-section of the coating layer of FBN nanosheets on the separator; **c.** high-magnification SEM image of the surface of the FBN separator; **d.** Raman spectrum of FBN nanosheets.

To evaluate the electrochemical performance of FBN separator, cycling performance was examined first. Figure 2a compares the cycling performance between two cells (sulfur areal density of $0.67 \pm 0.02 \text{ mg cm}^{-2}$) with and without FBN separators at a current density of 0.2C (335 mA g^{-1}).

The initial discharge capacity of the cell with the FBN separator is measured to be 1485 mA h g⁻¹, followed by a moderate drop until 10th cycle, which is related to the activation process in Li-S cells.¹⁵ A stable high discharge capacity of 1250 mA h g⁻¹ is achieved after 100 cycles. In contrast, a significant capacity decay is observed in the cell without FBN separator with a capacity retention of only 498 mA h g⁻¹ after 100 cycles. These results demonstrate that the coating of FBN nanosheets on Celgard separators can effectively prevent PS shuttle effect. To optimize the FBN coating, 5 separators with different FBN coating densities (0.05, 0.10, 0.16, 0.21 and 0.27 mg cm⁻²) were prepared by coating different amounts of FBN solution. Cycling performance at a current density of 1C of the cells with FBN separators of different coating thickness is shown Figure 2b. It shows that cycling stabilities are enhanced by increasing the FBN content in coating suspension, however, increasing the FBN content more than 0.21 mg cm⁻² doesn't have any positive effect on cycling stability. Hence, FBN coating with FBN nanosheet content of 0.21 mg cm⁻² is selected as a model system to further experimental work.

Figure 2c represents the rate capability performance between the cells with and without FBN separators. The consecutive cycling behavior at different charge-discharge rates were measured after 9 cycles in ascending steps from 0.2 to 7C and followed by a return to 0.2C up to total 60 cycles. The cell with a FBN separator exhibits a superior rate capability with a capacity retention of 1486, 1232, 1194, 1008, 904 and 718 mA h g⁻¹ at current densities of 0.2, 0.5, 1, 3, 5 and 7C, respectively, demonstrating a remarkably improved rate capability and reversibility. Furthermore, when the current density is returned to 0.2C, the capacity recovers to 1225 mA h g⁻¹ after 60 cycles (~83% retention of the capacity of 1486 mA h g⁻¹ measured at 0.2C initially), indicating good stability of the cell under high current charging and discharging conditions. On the contrary, the cell with a Celgard separator suffers from significant capacity decay at high current densities of

higher than 3C, which could be related to the severe PS shuttle effect.³⁶⁻³⁸ In addition, discharge-charge potential profiles obtained at different C-rates of the cell with an FBN separator for the selected cycles are shown in Figure 2d. Two distinct discharge plateaus, corresponding to the conversion of S to long-chain lithium PSs and the subsequent formation of solid-state discharge products of lithium sulfides, can be seen clearly. Similar plateaus are observed previously in Li-S cells with carbon-sulfur cathodes.^{39,40} Impressively, the long charge plateaus are still present at each current rate and even at a very high current rate of 7C, demonstrating stable discharge-charge potential profiles of the FBN separator cell.

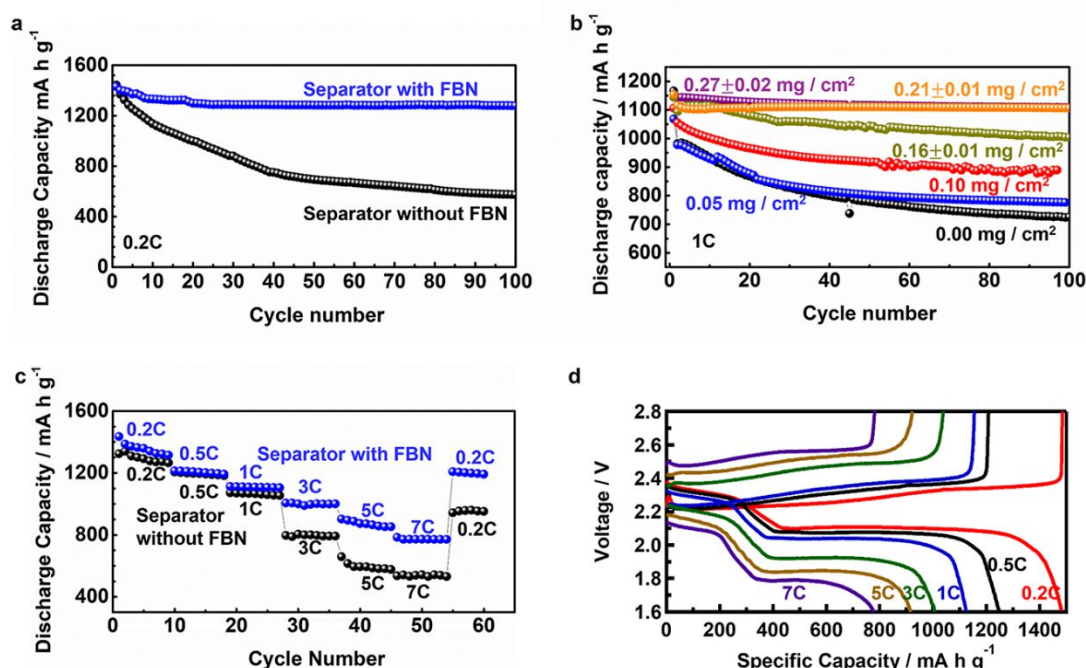


Figure 2 Electrochemical performance: **a.** cycling stability of two cells with and without FBN separator at 0.2C up to 100 cycles; **b.** cycling performance of the cells with FBN separators prepared using the FBN solution with different content of FBN nanosheets; **c.** rate capability of the cells with and without FBN; **d.** charge-discharge curves of the cells with FBN obtained at various current rates.

Figure 3a and b shows the long-term cycling performance of the cells. Figure 3a compares cycling stability of the cells with and without FBN separators at 1C and 3C. The cells with FBN separators display stable cycle life with retained capacities of 1027 mA h g⁻¹ at 1C and 897 mA h g⁻¹ at 3C after 1000 cycles with capacity retention of ~89% and ~83%, respectively. The capacity retention of ~89 and ~83% represent the slow decay rates of 0.011% per cycle at 1C (with respect to the initial capacity of 1154 mA h g⁻¹) and 0.017% per cycle at 3C (with respect to the initial capacity of 1090 mA h g⁻¹), corresponding to Coulombic efficiency of 99.3% at 1C and 99.8% at 3C, respectively (coulombic efficiencies curves can be found in Figure S2). In contrast, the initial discharge capacities of the cells with Celgard separators were measured to be 1166 mA h g⁻¹ at 1C and 1083 mA h g⁻¹ at 3C, followed by a huge drop of capacities with capacity retention of 456 mA h g⁻¹ (retention of only 39%) at 1C and 295 mA h g⁻¹ (retention of only 27%) at 3C, respectively. The enhanced cycling performance of the cells with FBN separators was further evaluated at a very high current rate of 7C. Figure 3b shows the cell is capable to tolerate such a high current density and the stable capacity was measured to be as high as 718 mA h g⁻¹ even after 1000 cycles with a capacity retention of ~91.5% and the coulombic efficiency of the cells with the FBN separator are over 98% during 1000 cycles, This excellent performance indicates that the incorporation of FBN nanosheets onto Celgard separators improves the electrochemical stability of the Li-S cell. Cyclic voltammetry (CV) was used to investigate the kinetics of electrochemical reactions in the cells with and without FBN separators after long term cycling. The CV curves for the cells after 1001 and 1002 cycles of discharge-charge at 3C are depicted in Figure 3c and d. The CV curves show two pairs of distinct and redox peaks for both electrodes. The reduction peaks at ~2.33 V and ~2.06 V are for the cells with FBN separators and peaks at ~2.35 V and ~2.08 V for the cell without FBN

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3 separators. These peaks show typical two-step sulfur reduction process which represents the
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5 conversion of elemental sulfur to long-chain polysulfides and long-chain polysulfides to short-
6
7 chain polysulfides.⁴⁶ The corresponding oxidation peaks at ~2.27 V and ~2.36 V for the cell
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9 with FBN separators and the peaks at ~2.31-2.33 V and ~2.36 V for the cell without FBN
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11 separators show the oxidation reactions from short-chain polysulfides to S_8/Li_2S_8 .⁴⁶ It is observed
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13 that the cell with FBN separators exhibits overlapping cathodic and anodic peaks while maintains
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15 their sharp shape and displays no obvious intensity change and potential shifts, indicating superior
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17 cycle stability and highly reversible redox reactions in the cell. In contrast, deformed and widened
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19 (with decreased integral areas) redox peaks are seen in the cell without FBN separators, which
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21 suggests a sluggish kinetic process. In order to confirm the PSs concentration on the anode side of
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23 the Li-S batteries with FBN separator, an H-type cell was prepared and cycled. Absorption of PSs
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25 was not found in UV spectrum (Figure S3).
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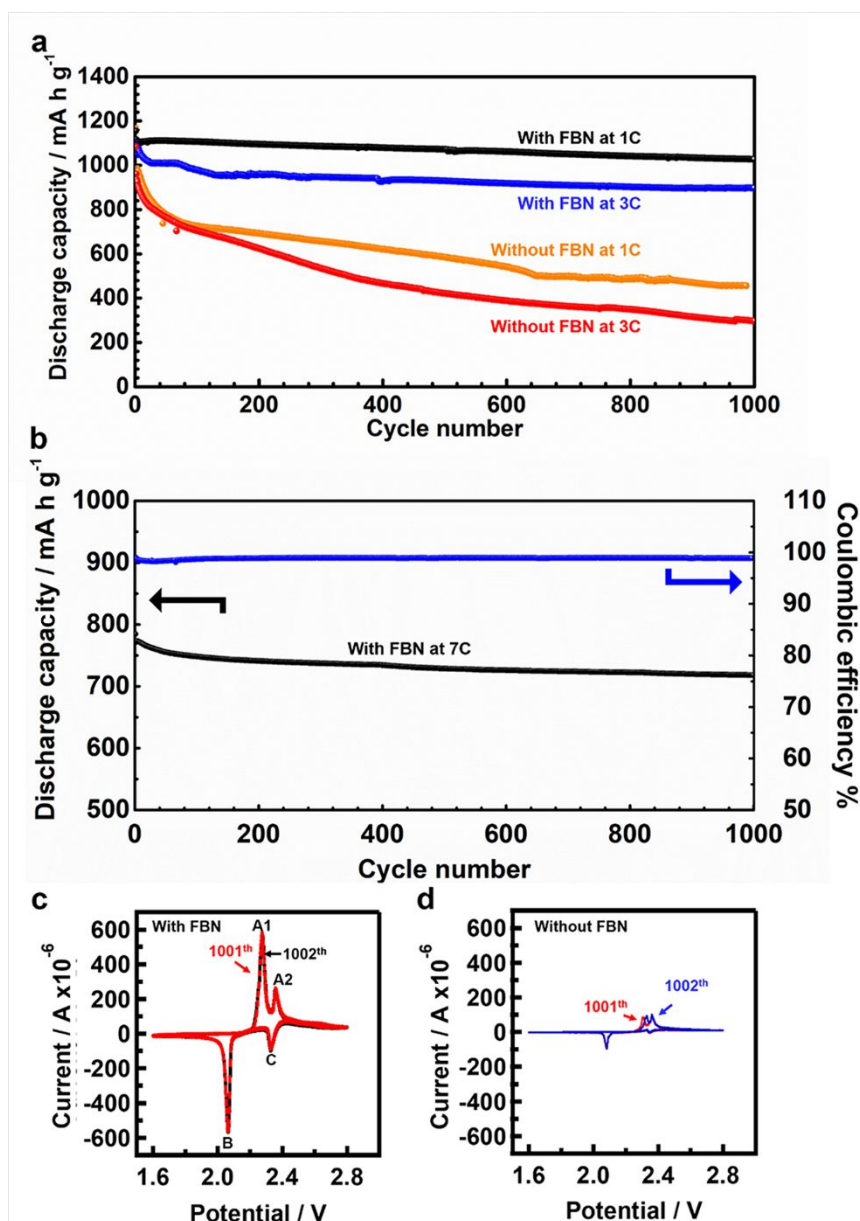


Figure 3 Long life cycling performance: **a.** cycling stability of the cells with and without FBN separators at 1C and 3C up to 1000 cycles; **b.** cycling stability of the cell with FBN separators at a high current rate of 7C up to 1000 cycles; and **c, d.** CVs of the cells with and without FBN separators after long-term cycling at 3C.

Influence of electrolyte to sulfur ratio has been investigated. The above results were recorded from the cells with 200 μL electrolyte. Half amount of electrolyte (100 μL) was used in some

cells, which gives an electrolyte/sulfur ratio of $83.3 \pm 0.6 \mu\text{L mg}^{-1}$. These cells were tested at 0.2C, 1C and 3C up to 100 cycles. Figure S4 shows all three cells performed a good stability and a high capacity of 1300 mA h g^{-1} at 0.2C. Under high current densities of 1C and 3C, the cells worked well but their capacities are lower than those of the cells with a high ratio of electrolyte to sulfur at $166.7 \pm 0.2 \mu\text{L mg}^{-1}$ because of less electrolyte available in the cells.

The cathodes of high areal densities of sulfur at 1 ± 0.2 and $2 \pm 0.5 \text{ mg cm}^{-2}$ were also tested with or without FBN separators. Figure S5 shows that the cells with the FBN separators have a better cycling performance than those without using FBN separators, which is in good agreement with the cells of sulfur areal density at 0.67 mg cm^{-2} (Figure 2a and Figure 3a). With FBN separators, at a high charging rate of 1C, the cells of 1 ± 0.2 and $2 \pm 0.5 \text{ mg cm}^{-2}$ have high capacities of 956 and 870 mA h g^{-1} , respectively.

To ascertain the interfacial behaviors of the electrodes with and without FBN separators, impedance spectra (EIS) were measured. The Nyquist plots of the cycled cells (first discharge state) are shown in Figure 4a, b. The Nyquist plots are composed of a depressed semicircle in the HF (high frequency) region, a depressed semicircle in MF (medium frequency) region, and a straight slopping line in the LF (low-frequency) region. To get a better understanding of the change of impedance parameters with and without FBN separators, the Nyquist plots were analyzed by Z-view software. Based on the characteristic of the Nyquist plots, we propose a simplified circuit for the EIS spectra, as presented inset of the Figure 4a. In the proposed circuit model, R_e is the resistance of electrolyte, $R_{\text{int}}//\text{CPE}_{\text{int}}$ is the interphase contact resistance and its related capacitance, $R_{\text{ct}}//\text{CPE}_{\text{ct}}$ is the charge-transfer resistance and its related capacitance, and CPE_{dif} is diffusion impedance that probably represents Li ions diffusion process. The first semicircle in the HF region is related to the interphase contact resistance and its related capacitance ($R_{\text{int}}//\text{CPE}_{\text{int}}$) in the sulfur

electrode bulk, which simulates the process of electron conduction from the current collector to reaction sites. The production of insulating reduction products during discharge on particle surface which could increase the interphase contact resistance and enlarge the HF semicircle moderately. The second semicircle in the MF region is related to the charge-transfer resistance and its related capacitance (R_{ct}/CPE_{ct}), which reflects the charge-transfer process at the interface between the conductive agent and the electrolyte.

From the Nyquist plots, the semicircle diameter of the cell with Celgard separator is smaller than that of the cell with FBN separator, implying the interphase contact resistance and charge-transfer resistance of the cell with FBN separator are higher than that of the cell with Celgard separator. The interphase contacts resistances (R_{int}) were calculated to be approximately 4.16 Ω for the cell with a Celgard separator and 24.4 Ω for the cell with a FBN separator after fitting with the equivalent circuit diagram in Figure 4a, respectively. The charge-transfer resistances (R_{ct}) were also very high for the cell with a FBN separator (88.7 Ω) whereas it was as low as 12 Ω for the cell with a Celgard separator. The higher interphase contact resistance in the cell with a FBN separator is attributed to the incorporation of insulating FBN nanosheets into the Celgard separator which directly touch the surface of sulfur cathode, and thus increases the contact resistance.^{47, 48, 49, 50} On the other hand, the higher R_{ct} value of the cell with a FBN separator confirms that the FBN separator is also capable to inhibit/prevent PS transfer and accumulate insulating discharge products of Li_2S and Li_2S_2 .⁵¹ These results demonstrate that the FBN nanosheets enhance the inhibition of PS transfer and limits internal redox shuttle.

To further demonstrate the ability of PSs rejection by FBN separator, an H-type cell with a changeable separator was employed (Figure 4c, d). The H-type cells were filled with yellow-colored PS solution (0.1 M) in the left-side chamber and a color-less electrolyte in the right-side

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3 chamber. After 24 hours, the H-type cell with a FBN separator shows no color change in the
4 electrolyte chamber (Figure 4c), whereas the electrolyte in the H-type cell with a Celgard separator
5 becomes yellow (Figure 4d), indicating a significant rejection of PSs by the FBN separator
6 compared to the Celgard separator. To confirm these findings, both separators (FBN and Celgard)
7 after cycling were collected and soaked in a color-less electrolyte solution for a certain period of
8 time. Remarkably, FBN separator displays no color change of electrolyte solution (compare to
9 Celgard), again confirming that FBN separator rejected negative charged PSs during cycling
10 process (Figure 4e). To further clarify the role of FBN nanosheets, 2 mg of FBN nanosheets was
11 added into a 0.001 M PS solution, and the PS solution was separated from the FBN nanosheets
12 using centrifugation treatment at 7200 rpm for 30 min. The 0.001M PS solution and the collected
13 PS solution were tested for ultraviolet absorption (UV) tests. As shown in Figure S6, the peak of
14 PS ions is present even after mixing with FBN, which indicate no adsorption of PS ions on the
15 FBN nanosheets because of the attached CO_3^{2-} groups.
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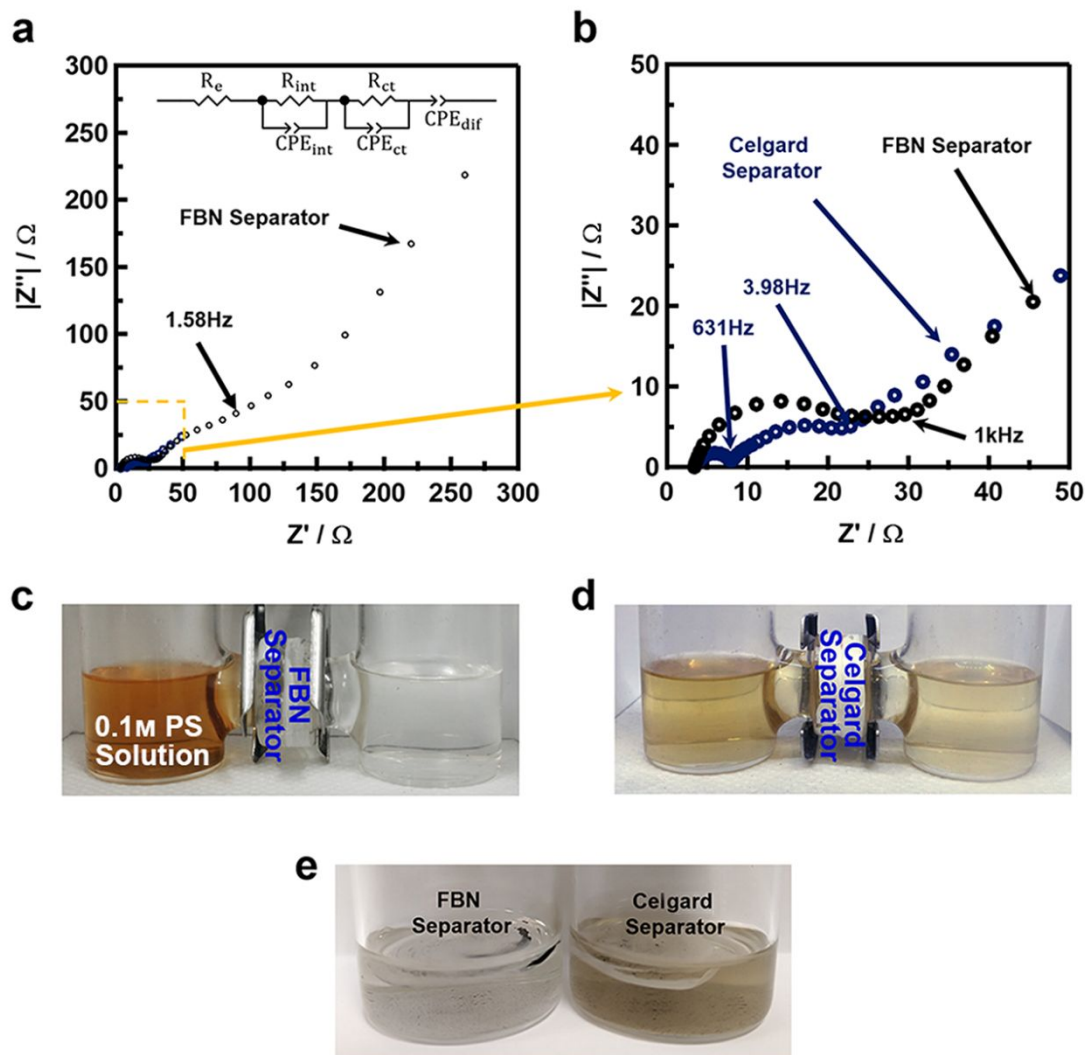


Figure 4 Diffusion ability and inner conductivity of Cells with different separator **a**, **b**. Electrochemical impedance spectroscopy (equivalent circuit model is depicted inset of Figure 4a) of the cells with and without FBN separators at the end of first discharge; **c**. color contrast of PS solution in the left-side chambers and color-less electrolyte in the right-side chamber in a H-type cell with FBN separator after 24 hours; **d**. color contrast of PS solution and color-less electrolyte in a H-type cell with Celgard separator after 24 hours; and **e**. color contrast of the separators (soaking in a color-less electrolyte) with and without FBN collected from cycled cells.

As discussed above, the cells with FBN separators exhibit excellent electrochemical performance due to the rejection of migration of PSs through FBN separator. On the other hand, Li ions permeability of the cell with FBN separators is also important parameter for its performance in Li-S cells, especially for excellent rate capability. Hence, Li ions diffusion coefficient was calculated using the Randles-Sevick equation with a series of CV measurement as shown in Figure 3c and Figure 5a, b.^{12, 52, 53}

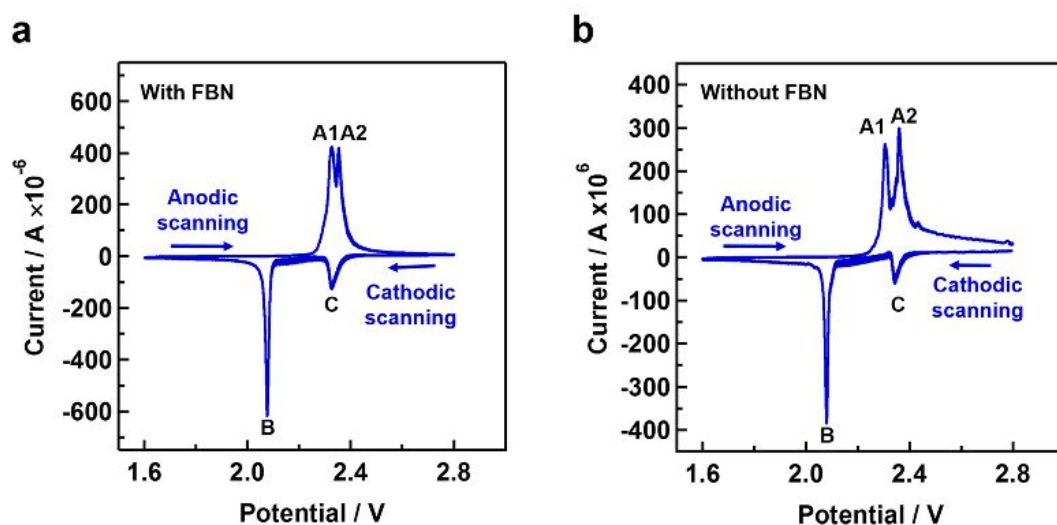


Figure 5 CVs for the first cycle at a scan rate of 0.01mVs⁻¹: **a.** cell with a FBN separator and **b.** cell with a Celgard separator (without FBN).

The diffusion coefficient of Li ions, D_{Li^+} (cm² s⁻¹), was evaluated by cyclic voltammetry and calculated according to the Randles-Sevick equation: $I_p = 2.69 \times 10^5 n^{1.5} A D_{Li^+}^{0.5} C_{Li} v^{0.5}$ (1)^{12,52,53} in which I_p (A) is the peak current, n signifies the number of electrons in the reaction (for Li-S cells, $n = 2$), A (cm²) designates the electrode area (1.538 cm²), C_{Li} (mol mL⁻¹) means the Li ions concentration

in the electrolyte, and v attitudes to the scanning rate ($V s^{-1}$). The CV measurements with different scanning speeds are plotted in Figure S7a, b in order to indicate a diffusion-controlled process.

Li ions diffusion coefficient, D_{Li^+} ($cm^2 s^{-1}$), is used to compare the transference number of Li ions according to the equation: $t_{Li^+} = \frac{D_{Li^+}}{D_{Li^+} + D_-}(2)$,⁵⁴ in which t_{Li^+} is the transference number of Li ions, D_- is the average diffusion coefficient of anion (including TFSI⁻ and S_x^{2-}) in the electrolyte. The transference number of the whole cell accorded to a standard calculation equation: $t_{\pm} = \frac{z_{\pm} \Delta n_a}{Q/F}(3)$, in which $t_{\pm}$ is the transference number of Li ion (+) and other anion (-), $z_{\pm}$ is the charge number (here, $z_+ = 1$ and $z_- = 1$ or 2), Δn_a is the transported amount of substance of ion (mol) including Li ions and other anion, Q is the quantity of electricity which passed through the whole cell (C), F is Faraday constant.

In this study, the equations $t_{Li^+} = \frac{D_{Li^+}}{D_{Li^+} + D_-}$, $t_{Li^+} = \frac{z_+ \Delta n_{Li^+}}{Q/F}$ and $t_- = \frac{z_- \Delta n_-}{Q/F}$ were employed to explain anion diffusion process. Here, $D_- = D_{S_x^{2-}} + D_{TFSI^-}$, $\Delta n_- = \Delta n_{S_x^{2-}} + \Delta n_{TFSI^-}$. Q does not change during charge-discharge processes at the same current density in different cells with and without FBN separators. The increase in D_{Li^+} leads to an increase of Δn_{Li^+} and t_{Li^+} . Similarly, the decrease of transference number of anion ($t_{Li^+} + t_- = 1$) leads a lower transported amount of substance in one-unit time which can be accorded to the definition of diffusion coefficient. The cathodic and anodic peaks in the CV curves (Figure 3c and Figure 5a, b) are assigned as B, C and A1, A2, respectively. Hence, the diffusion coefficients of Li ions with FBN separator at different CV voltage regions can be determined according to the equation (1) (Randles-Sevcik equation): $D(A1)_{FBN} = 1.22 \times 10^{-8}$, $D(A2)_{FBN} = 1.22 \times 10^{-8}$, $D(B)_{FBN} = 2.71 \times 10^{-8}$ and $D(C)_{FBN} = 8.82 \times 10^{-10} cm^2 s^{-1}$ for the first cycle. Similarly, the calculated diffusion coefficients of Li ions with a Celgard separator (without FBN) according to the equation

(1) is: $D(A1)_C = 6.56 \times 10^{-9}$, $D(A2)_C = 6.13 \times 10^{-9}$, $D(B)_C = 9.45 \times 10^{-9}$ and $D(C)_C = 3.57 \times 10^{-10}$ cm s^{-1} for the first cycle. The calculated values of diffusion coefficients of Li ions of the cell with an FBN separator are three times higher than those of the cell with a Celgard separator (without FBN) during the first discharge process. The diffusion coefficients of Li ions in the cell with FBN separator at the 1001th cycle are $D(A1) = 2.62 \times 10^{-8}$, $D(A2) = 3.21 \times 10^{-9}$, $D(B) = 2.54 \times 10^{-8}$ and $D(C) = 1.05 \times 10^{-9}$ cm s^{-1} . One can see that the diffusion of Li ions in the cell with a FBN separator is quite similar at the first discharge and even after 1000 cycles.

The diffusion coefficients of Li ions calculated from the peaks B and C indicate the performance of the ion transfer situation and internal redox shuttle process of polysulfide during discharging process.³⁸ It suggests that the Li ion transfer situation at the whole recycling process is stable in the cell with FBN separators and the element constitute of both anode and cathode are stable. These results demonstrate that the diffusion speed of Li^+ in the cell with a FBN separator is much higher than that in the cell without FBN separator. However, this high Li ion conductivity behavior, which leads to the higher rate performance of the Li-S cells with the FBN separator, is different from the EIS results because EIS cannot detect the reaction occurring faster or slower than the testing time and it is difficult to separate processes occurring on the same timescales.^{55,56} Xie et al. also speculated the D_{Li} values obtained by EIS were not reliable at the two-phase transition reaction range due to the dynamic process between two-phase during the diffusion,⁵⁷ such as S_8 (solid) to S_6^{2-} (liquid) in this research. Moreover, the transference number, t_{Li^+} , in the cell with a FBN separator is higher than that of the cell without FBN separator because FBN is recognized as a Lewis base which can enhance the transference number,⁵⁸⁻⁶¹ indicating that the diffusion of anions such as $D_{\text{S}_x^{2-}}$ and D_{TFSI^-} decreases because of the identities $t_+ + t_- = 1$. The overall

electrochemical performance (capacity, cycling stability and rate capability) of the Li-S cell with a FBN separator is much better than the cell without FBN separator.

Conclusions

A multifunctional separator is developed for Li-S batteries by coating a commercial Celgard separator with a thin layer of FBN nanosheets. This BN separator overcomes internal redox shuttle which is one of the major constraints associated with the traditional separator. Moreover, the incorporation of FBN nanosheets into Celgard separators for the Li-S cells is afforded an improved capacity from 456 mA h g⁻¹ (Celgard, without FBN) to 1027 mA h g⁻¹ (with FBN) at 1C and 295 mA h g⁻¹ (Celgard, without FBN) to 897 mA h g⁻¹ (with FBN) at 3C after 1000 cycles with negligible capacity decay rates. Most importantly, the cells with FBN separators show an ultra-stable capacity of 718 mA h g⁻¹ with a capacity retention of ~91.5% at a high rate of 7C over 1000 cycles, which might be related to the increased Li ion diffusion coefficient of the cell as well as rejection of PSs. It is demonstrated that this BN separator can restrict dissolved PSs migration through the separator due to strong electrostatic interaction between negatively charged PSs products and FBN nanosheets, which enhances ion transfer and inhibits internal shuttle effect, offering the cells with a high capacity, high rate capability and long-term cycling stability. Hence, our BN separators would be a promising strategy for the development of high-performance Li-S batteries.

Experimental

Preparation of boron nitride nanosheets: The preparation method of the boron nitride nanosheets is following the approach which has been published in a previous report.²⁷ 0.28 g boron trioxide and 1.95 g guanidine hydrochloride were mixed in 20 ml methanol to form a clear, colorless solution. After 24 h of fast stirring a white crystalline powder was formed. The crystalline

powders were heated to 1100 ° for 2 h in a nitrogen-15% hydrogen gas flow. White porous BN nanosheets were collected after cooling to room temperature.

Functionalization of boron nitride nanosheets: To achieve functionalized BN nanosheets with CO_3^{2-} groups, BN nanosheets and guanidine carbonate (Sigma-Aldrich) with a mass ratio of 1:1 were ball-milled under nitrogen atmosphere for a period of 24 hours.³⁰ The milled product was collected and washed with deionized water and lyophilized. The final product is denoted as FBNs.

Fabrication of FBN separation: A stock solution with a mixture of polyvinylidene fluoride (PVDF) (1 wt.%) and N-Methyl-2-pyrrolidone (NMP) was prepared. 100 ml stock solution was used to disperse 10 mg of FBN. Controlled amount of this dispersion was poured onto a Celgard 2400 separator to give different coating density, and the coating layer was formed with the help of a vacuum filtration system. Huge amount of deionized water was used to wash out extra PVDF and NMP. This FBN coated separator was then dried in a vacuum oven at 60°C for 24 hours. To achieve different coating thickness, five different coating suspensions were prepared by varying the weight of FBN.

Materials characterization: Cross-section and surface morphology of the FBN coated separator were examined by scanning electron microscopy operated at 3 kV (Hitachi S4500 Zeiss Supra 55VP). To achieve enough conductivity, the FBN separator was coated with carbon and gold before SEM analysis. The functional group of boron nitride nanosheets was confirmed by Raman spectroscopic characterization using a Renishaw *in Via* micro-Raman system equipped with a 514.5 nm laser. Zeta potential measurement was carried out in ethanol solution by using Malvern ZETASIZER Nano series.

Electrochemical characterization: The sandwich-like Graphene/Sulfur electrode was prepared with a mixture of 12 wt.% high porous graphene (graphene-supermarket, USA), 8 wt.% high

surface graphene (graphene-supermarket, USA) and 80 wt.% sulfur. The mixture was hand grounded for 2 hours and followed by a heat treatment at 300°C in an airtight container for 24 hours. The heat-treated sample was mixed with Super P Li (Timcal Ltd.) and PVDF (in a weight ratio of 80:15:5) in NMP solution and stirred for 24 hours to produce a homogeneous slurry. The slurry was then coated onto a carbon coated aluminum foil substrate and heated in an air oven at 60 °C for 24 hours. These base electrodes were furthered coated with a graphene layer prepared by a mixture of graphene and PVDF (in a weight ratio of 95:5) in NMP solution.⁹ The electrodes were further dried in an air oven at 60°C for 24 hours. All Graphene/Sulfur electrodes were cut into a circular size. The sulfur loading in the cathode is about 63 wt.% of total weight (including sulfur, carbon, and PVDF binder).

CR2025 coin-type half cells were assembled inside the Ar-filled glove box (Innovative Technology, USA) with Graphene/Sulfur as working electrode and Li metal as counter electrode. The electrolyte was 1 M Bis(trifluoromethane)sulfonimide lithium salt (LiTFSI; Sigma Aldrich) in a 1:1 (by volume) mixture of 1, 2-dimethoxyethane (DME; Sigma Aldrich) and 1, 3-dioxolane (DOL; Sigma Aldrich). A microporous polypropylene (Celgard 2400) and FBN coated polypropylene used as separators. The amount of the electrolyte in a cell is 200 μ L and the thickness of the Li anode is around 0.2 mm. The Li-S cells were galvanostatically discharged-charged between 1.6-2.8 V at different current densities using a LAND battery tester (Wuhan LAND Corporation, China). Cyclic voltammograms (CVs) were recorded using Solartron Analytical electrochemical workstation at the scan rate of 0.01, 0.02, 0.03 and 0.04 mV s⁻¹. Electrochemical impedance spectroscopy (EIS) was performed on the cells over the frequency range of 0.01 Hz to 1 MHz with an AC amplitude of 5 mV using Solartron computer-controlled electrochemical analyzer.

ASSOCIATED CONTENT

Supporting Information. Coulombic efficiencies of each kind of Li-S cells at different current rate, Ultraviolet spectrum of PS solution before and after soaking the FBN powder and performance of the cells with different electrolyte and sulfur loading. (PDF)

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Author Contributions

Y.F., W.W.L. and Y.C. conceived the project. Y.F., W.W.L. and Y.C. designed the experiments. Y.F., W.W.L., and D.L. carried out the experimental works. Y.F., T.T. analysed electrochemical experiments. Y.F., J.M.W., C.Y. performed materials characterization. Y.F., W.W.L. M. M. R. and Y.C. wrote the manuscript. All authors contributed to analyses the data and discussions regarding the research.

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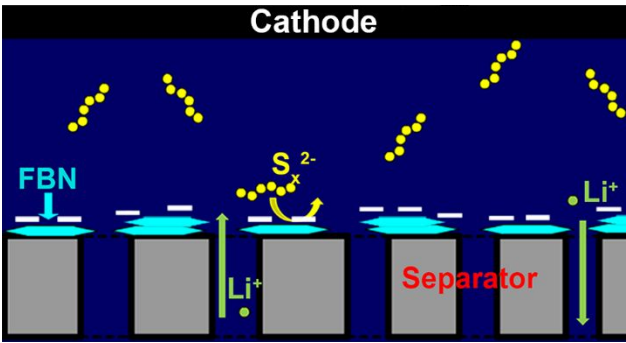
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“Boron nitride separators prevent polysulfide migration through the separator through ion-repelling effect of negatively-charged boron nitride nanosheets.”



Graphic Table of Contents

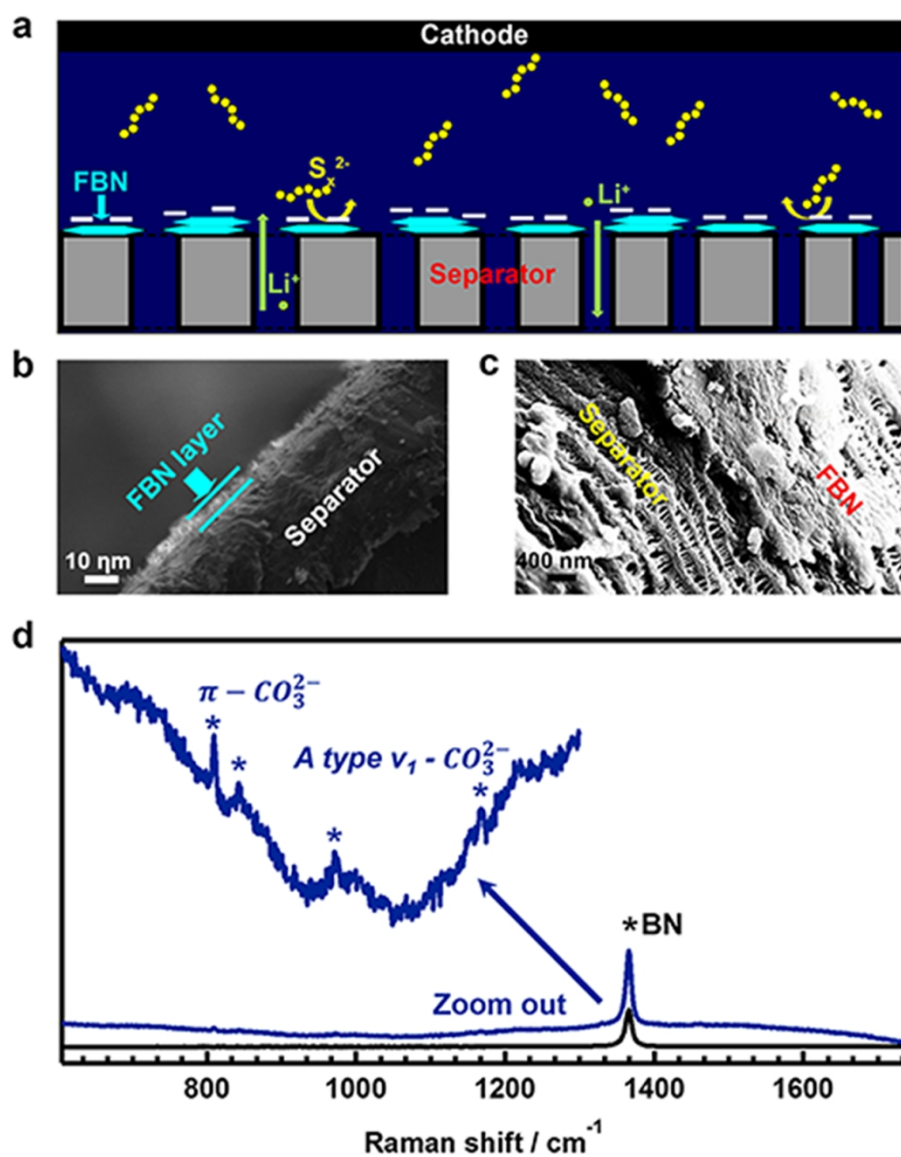


Figure 1 Structure of the Li-S cell with an FBN separator. a. schematic configuration of a sandwich-like Li-S cell with a FBN separator; b. SEM image of the cross-section of the coating layer of FBN nanosheets on the separator; c. high-magnification SEM image of the surface of the FBN separator; d. Raman spectrum of FBN nanosheets.

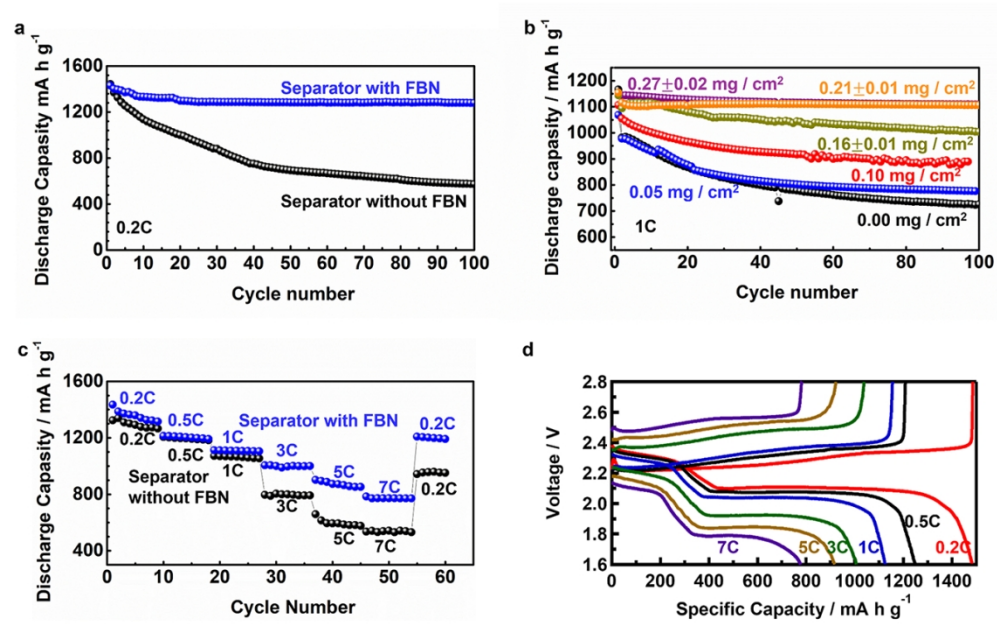


Figure 2 Electrochemical performance: a. cycling stability of two cells with and without FBN separator at 0.2C up to 100 cycles; b. cycling performance of the cells with FBN separators prepared using the FBN solution with different content of FBN nanosheets; c. rate capability of the cells with and without FBN; d. charge-discharge curves of the cells with FBN obtained at various current rates.

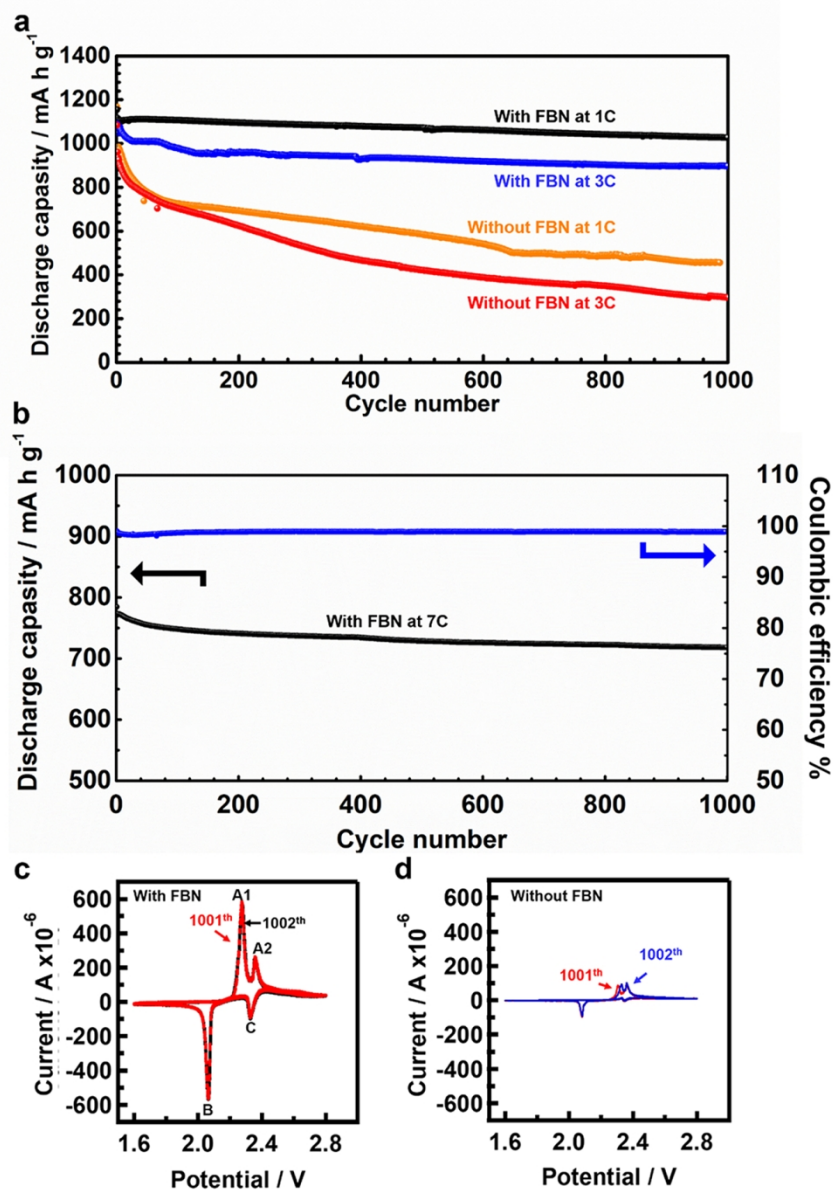


Figure 3 Long life cycling performance: a. cycling stability of the cells with and without FBN separators at 1C and 3C up to 1000 cycles; b. cycling stability of the cell with FBN separators at a high current rate of 7C up to 1000 cycles; and c, d. CVs of the cells with and without FBN separators after long-term cycling at 3C.

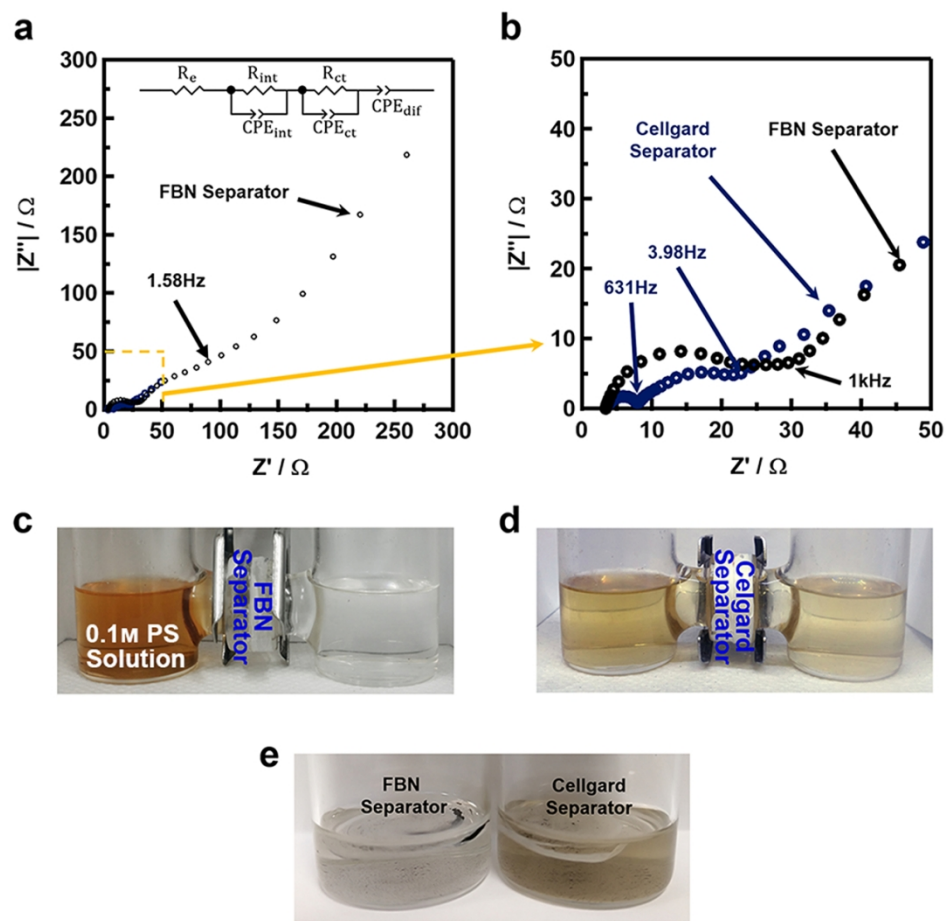


Figure 4 Diffusion ability and inner conductivity of Cells with different separator a, b. Electrochemical impedance spectroscopy (equivalent circuit model is depicted inset of Fig. 4a) of the cells with and without FBN separators at the end of first discharge; c. color contrast of PS solution in the left-side chambers and color-less electrolyte in the right-side chamber in a H-type cell with FBN separator after 24 hours; d. color contrast of PS solution and color-less electrolyte in a H-type cell with Celgard separator after 24 hours; and e. color contrast of the separators (soaking in a color-less electrolyte) with and without FBN collected from cycled cells.

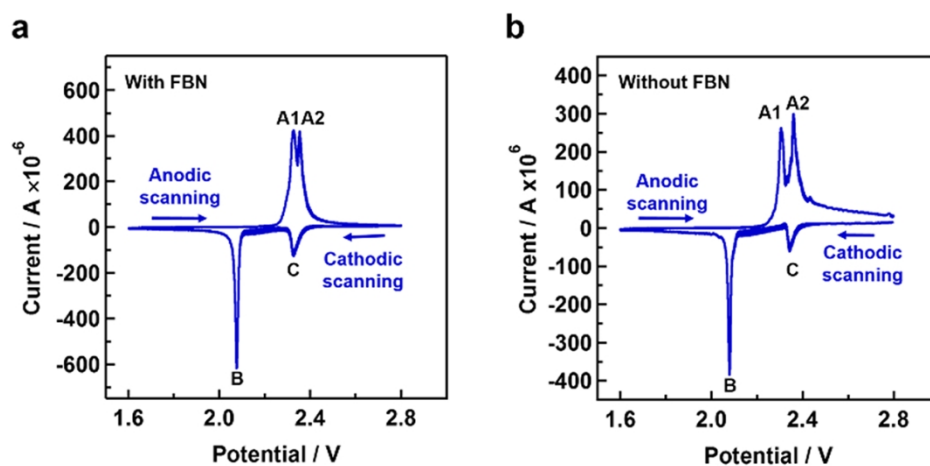


Figure 5 CVs for the first cycle at a scan rate of 0.01mVs⁻¹: a. cell with a FBN separator and b. cell with a Celgard separator (without FBN).